MOHAMMAD AKIF AKHTAR

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AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | NLP | Model Deployment

PROFESSIONAL SUMMARY

AI/ML Engineer skilled in building and deploying Machine Learning and Deep Learning solutions across Computer Vision and NLP, including CNN and YOLO-based image classification, TF-IDF/NLP text models, unsupervised learning (clustering, market basket analysis), and Flask REST APIs served in production with Gunicorn on Render. Experienced in the full pipeline from data preprocessing and model training to evaluation and deployment.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Data Structures & Algorithms (DSA)
Machine Learning: Scikit-learn, Random Forest, Logistic Regression, SVM, Decision Tree, Classification, Clustering (K-Means), Feature Engineering, Model Evaluation, Cross-Validation
Deep Learning: TensorFlow, Keras, CNN, YOLO11 (Ultralytics), Neural Networks, Data Augmentation
Computer Vision: Image Classification, Image Preprocessing, OpenCV, Pillow
NLP: TF-IDF, Text Classification, Text Preprocessing
Analytics & Statistics: Statistical Analysis, RFM Analysis, K-Means Clustering, Market Basket Analysis (Apriori), Association Rules
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Excel
APIs & Deployment: Flask, REST APIs, Gunicorn, Render, Model Serialization (Pickle/Joblib), ReportLab
Tools: Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

RetinaVision AI — Diabetic Retinopathy Detection | *YOLO11, Flask, OpenCV, ReportLab*
- Built a YOLO11-based computer vision system to classify diabetic retinopathy severity across 5 clinical stages, achieving 84.97% Top-1 accuracy and 100% Top-5 accuracy on 366 evaluation images.
- Developed a Flask-based inference pipeline with OpenCV/Pillow preprocessing, Top-3 confidence-based predictions, and automated PDF report generation using ReportLab.

Loan Approval Prediction — Machine Learning | *Scikit-learn, Random Forest, Flask, Render*
- Built a Random Forest classifier to predict loan approval, achieving 96.8% accuracy, 0.96 F1-score, and 0.97 ROC-AUC, outperforming Logistic Regression and Decision Tree baselines.
- Serialized the model with Joblib and deployed a Flask web app live on Render for real-time predictions.

Customer Segmentation — Unsupervised Learning | *Scikit-learn, K-Means, Apriori*
- Performed RFM (Recency, Frequency, Monetary) analysis on e-commerce transaction data and applied K-Means clustering to segment customers into 4 actionable groups.
- Applied Market Basket Analysis using the Apriori algorithm to mine frequent itemsets and generate association rules (support, confidence, lift).

Leaf Disease Prediction — Deep Learning | *TensorFlow, Keras, CNN*
- Trained a custom CNN from scratch using TensorFlow/Keras to classify 15 leaf disease classes across Bell Pepper, Potato, and Tomato crops, achieving 96.06% validation accuracy and 95.04% macro F1-score.
- Applied data augmentation (flip, rotation, zoom, contrast) and image preprocessing, using EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau to optimize training.

Email Spam Classifier — NLP | *TF-IDF, SVM, Scikit-learn*
- Built an NLP pipeline (text cleaning, stopword removal, stemming/lemmatization, TF-IDF vectorization) to classify emails as spam or legitimate, evaluating Logistic Regression, Naive Bayes, SVM, and Random Forest.
- Selected SVM as the top performer (99.21% accuracy) and built a Flask web interface returning spam probability and risk level in real time.

EXPERIENCE

Samatrix Consulting Private Limited — ML/Data Analysis Intern 01 Jul 2026 – 28 Jul 2026
- Performed Python-based financial data preprocessing, statistical analysis, and visualization for quantitative portfolio risk assessment.
- Calculated 1-Day 95% Value at Risk (VaR) to evaluate potential portfolio losses and support risk-based decision-making.
- Applied data analysis and machine learning techniques to financial datasets to identify patterns and generate actionable insights.

ACHIEVEMENTS

- Led a team to win the Best Visionary Team Award at an intra-university hackathon for DisasterSense, an AI-based disaster response/prediction project.

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML), Quantum University 2024 – 2028
CGPA: 8.25/10